



WriterINet: a multi-path deep CNN for offline text-independent writer identification

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Abstract

Handwriting-based identification is a fundamental pattern recognition problem that has attracted considerable interest in recent years. Writer identification is a major challenge, considering how diverse the written content is and how much handwriting differs between writers. This paper presents *WriterINet*, a CNN-based approach to learning and characterizing each writer's writing style. The proposed *WriterINet* approach takes handwritten documents as input, decomposing each of them into word images and connected component sub-images. Each segmented image is then fed into our feature learning step to generate discriminative and deep features. To this end, we developed a powerful deep CNN feature architecture consisting of two CNN streams derived from fine-tuning the ResNet-50 and DenseNet-201 models. Deep learned features are computed from all segmented images representing the writer's document and classified using a proposed 1D artificial neural network for predicting writer identification by averaging the similarity scores. Experimental results on IAM, ICDAR2013, CVL, IFN/ENIT, ICDAR2011, Firemaker and CERUG show that our *WriterINet* approach achieves the highest or competitive performance over the state-of-the-art.

Keywords Handwriting · Deep learning · Text-independent · Convolutional neural network · Off-line writer identification · Feature learning · 1D artificial neural network

1 Introduction

Handwriting is one of the most reliable behavioral biometric data for characterizing the individuality of the writer. It has the same importance as fingerprint systems for biometric identification. Unlike machine texts, handwriting contains distinguishing characteristics of the person who wrote it. The handwriting of the same person is often influenced by many

factors such as age, schooling, mood, time, and so on. This is called intraclass variance. The dissimilarity between samples from two different people is called interclass variance. A good biometric writing trace has a small intraclass variance and a large interclass variance. This means that a person will never produce the same writing style in exactly the same way twice. This makes handwriting analysis a challenging task and an interesting field of research for psychologists, graphologists, forensic scientists, and historians. It has been used for many centuries as a means of identity verification and pattern recognition.

Handwriting is an important aspect of document understanding and analysis with many scientific and technical locks. Existing approaches from the fields of pattern recognition and authentication have focused on developing new and reliable systems for handwriting analysis. These systems can be learned and applied to different writing styles with different application requirements. Handwriting analysis using artificial intelligence has many applications in modern life. It includes in full extension a number of applications such as classification of old documents [1], smart meeting rooms [2], online/offline verification of handwritten signatures [3],

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handwritten musical scores [4–6], and online/offline writer identification [7–9]. It also finds application in determining the writer's gender, age, and handedness [10], in forensics for identifying those responsible for fraudulent/threatening letters, ransom note, business analysis agreements [11], etc.

This paper deals with the problem of automatic identification of writers from handwritten images. It aims at recognizing a person by his/her handwriting. The goal is to effectively capture the variability between and within a writer. To achieve this, a high level of expertise is required to characterize such extreme variations. Writer identification is widely used in crime detection and historical document analysis [12]. It has attracted a great attention in recent decades because of the urgent need to develop biometric systems for many security applications.

Writer identification consists of matching a particular writing sample with reference writings whose writer is known. This matching is done using a similarity index, preceded by preprocessing of the handwriting image and feature extraction. There are still challenges in this research area, as documents have very complex writing styles with different languages. This work aims to help solve the general problems encountered at these different stages. The tasks relevant to writer recognition are verification, identification, and retrieval. The first task performs a one-to-one comparison to match an unknown writer with those given by the verification process [13,14]. This determines whether or not two queried handwriting samples are from the same person. When identifying writers, the system returns a list of likely writers, ordered by class, whose writing style is identical to the test sample. This is called a multi-class discrimination task, which performs a “one-to-many” search technique within a large handwritten database. When retrieving and searching for writers, the system retrieves from large database all document samples written by a particular scribe. This is neither a “writer identification” task, since the system outputs a set of documents rather than the names of the scribes, nor a “document retrieval” task. In this case, the classification criterion is the name of the writer. Writer verification, identification, and retrieval, all of which rely on pattern recognition and machine learning techniques to characterize the variability of writing. Writer identification is one of the most challenging tasks for the computer vision community. One of the interests that sustain the continued study of this task is that human intervention is unnecessary. Older forensic techniques are cumbersome to compare, match, and identify the authenticity of documents. Today, computer-based learning systems simplify and improve the identification task, reducing the search space for large databases. Currently, there are many public benchmarks with different scripts and languages used to evaluate new algorithms in this direction.

Research on writer identification can be divided into two groups: online and offline. Online methods deal with dynamic

features of writing such as stroke sequence, trajectory, height, writing speed and time. This temporal and spatial information is extracted in real time at the pen tip by special acquisition devices (e.g., Anoto pen and handheld electronic devices). The output, a sequence of signals, is then transmitted to computers for analysis. Offline methods view the writing samples as scanned input images that exhibit pattern variation. In this way, various image processing and segmentation techniques can be performed to improve recognition performance. Compared to the online method, the offline method is the focus of interest in the computer vision community, which has practical applications in security/biometric verification and forensic/historical document examination [15].

Depending on how the writing is generated, offline approaches to writer identification are classified into text-dependent and text-independent techniques. The first technique uses the same fixed handwritten text in both training and test sets. Although this strategy facilitates classification, it is not applicable in many practical cases due to the constraints imposed on the text content. In contrast, text-independent methods are not tied to the content of the text and are therefore more widely applicable. Since the style is visually distinguishable between writers, any specific text can be considered in experiments to identify query writers. This makes the text-independent method more challenging, as it must deal with image pixels with arbitrary fonts and different character shapes.

The presented paper addresses offline text-independent writer identification of handwritten texts using deep learning. Our proposed deep learning-based approach, called *Writer-INet*, consists of three steps. First, the written documents are submitted to the preprocessing and segmentation phase, where each written sample is decomposed into word images and connected sub-images. Second, a powerful deep CNN model is built from ResNet-50 [16] and DenseNet-201 [17] models (pre-trained on the training set of IAM database) for feature extraction. The proposed model extracts discriminative features of the input word and its associated image components. Third, a simple artificial neural network model (ANN) is developed to classify the obtained local features (corresponding to the words and the connected components). In this way, for each test document, the trained model predicts all the similarity values of its local feature vectors, based on which the writer identification is performed. The proposed framework is evaluated on seven well-known databases (Arabic IFN/ENIT, English ICDAR2013, English CVL, ICDAR2011 in hybrid languages, English IAM, Dutch Firemaker, and English-Chinese CERUG). Experimental results show competitive or high performance of the proposed system compared to the current SOTA. The main contributions of the present work are listed below:

- A novel and effective multi-path deep CNN-based approach, called *WriterNet*, for offline text-independent writer identification is proposed.
- For the first time, extensive experiments are conducted on nine challenging handwritten benchmarks in different languages (English, Arabic, French, German, Chinese, Dutch, Greek, and hybrid language).
- For the first time, a user-defined 1D artificial neural network (1D- ANN) is proposed in the classification phase (1D- ANN is basically used in signal processing for 1D data).
- The proposed *WriterNet* achieves competitive or highest SOTA performance on all benchmarks studied.

The remainder of the paper is organized as follows. Section 2 reviews recent research on offline text-independent writer identification. Section 3 describes the main steps of the proposed *WriterNet* framework. Section 4 describes the evaluated databases with their setup, and the experimental results are reported and analyzed together with those of SOTA. Finally, Sect. 5 concludes the paper and gives an outlook on future work.

2 Related works

Several survey papers [18,19] have dealt extensively with writer identification SOTA. The interest is to compare, evaluate, and develop reliable, near real-time, and robust approaches that provide high identification performance. In this area, abstract and discriminative writing style features need to be computed and details that reflect personal writing habits need to be extracted. This is a major challenge because of the sensitivity of the shape of the characters and the wide variation between writers. Dealing with such extreme variations greatly improves the identification task and reduces misclassifications. Most existing work generally considers the following pipeline: Preprocessing, Feature Extraction, and Classification. Here, we summarize the research work on writer identification. These works can be classified into four groups: Texture-based, Grapheme-based, Contour-based, and Auto-learning methods.

Texture-based methods consider the handwritten pattern as a texture image and extract features within blocks, regions, fragments, words, lines of text, or the entire handwritten image [20–22]. Local binary patterns (LBP), local ternary patterns (LTP), and local phase quantization (LPQ) have been used as global features in several recent works for writer identification [22–24]. In [7], LPQ, LBP and LTP methods have shown impressive results to cope with writing variability. They have been used as local features for connected components, based on which many-to-many feature matching is

performed to identify the writer. Similarly, in our previous work in [25], we proposed a new and effective feature method called Cross multi-scale Locally encoded Gradient Patterns (CLGP). This method captures texture information by constructing transform features and encodes the obtained texture codes within non-overlapping blocks at multiple scales via the histogram of oriented gradients (HOG). [20] presented a global approach based on texture analysis. The authors used the gray level run length matrix (GLRL) and gray level co-occurrence matrix (GLCM) to extract texture features, and applied the standard nearest neighbor rule for classification. Similarly, [21] constructed texture features computed using the Gray-Scale Co-occurrence Matrix (GSCM) and the multichannel Gabor filtering (MGF) technique. These features are used with weighted Euclidean distances (WED) and (K-NN) classifiers to identify writers. The run lengths of the hand-crafted descriptors were also examined. The run length of the LBP was used by [26] as a curvature-free feature for irregularly curved handwriting. Recently, Bahram [27] performed writer identification using texture features obtained from preprocessed partial images of handwriting. These include modified local binary pattern (MLBP) and ink-trace width and shape letters (IWSL) descriptors. Query writers were classified using the nearest neighbor rule based on the chi-square distance.

Contour-based approaches directly capture features in the contours of the handwriting and require segmentation of the text image to improve identification of the writer. The best known system using contour-based features is the one presented in [28]. The authors proposed a contour-based feature, the hinge feature. It computes the orientation and curvature information to provide a compact characterization of the individual handwriting style. Comparable results to the Hinge feature have been reported by [29]. Their system relies mainly on two different aspects of handwriting: the presence of redundant patterns in the handwriting and the extraction of its two visual attributes (orientation and curvature). Reference [30] proposed an accurate method based on pixel contours, Quill and QuillHinge features. It consists of a probability distribution of the relationship between ink direction and ink width. Reference [31] presented a method for extracting rotation invariant features from handwriting samples for writer identification. They incorporated the derivation between multiple points along the ink contours to improve the characterization of the writer. In the same way, [26] introduced the cloud of line distribution to characterize an irregularly curved handwriting.

Grapheme-based approaches capture the writing distribution of small traces called graphemes. These writing units can be encoded as local descriptors using encoding methods such as the bag-of-words approach (via a prototypical codebook generated by clustering). Reference [32] proposed an ensemble of grapheme codebook features for identifying writers.

Reference [8] used the beta-elliptic model to generate a synthetic codebook for feature extraction. They stated that this work was the first to use the concept of model-based synthetic codebooks for writer identification and verification. Ghiasi et al. [33] proposed an efficient codebook approach based on curve fragment and line fragment extraction to characterize writing style. Reference [34] presented a generic method for detecting junctions in handwritten samples. They used the detected transitions to identify the writer using junkets. The same authors proposed local contour fragments (kCF) and stroke fragments (kSF) in a later work [35]. Ahmed et al. [36] used the concept of a bag of features to train a k-means clustering algorithm that generates codebook features. In the classification phase, [36] applied the nearest neighbor and support vector machine methods to identify the queried handwritten documents. The scale invariant feature transformation descriptor (SIFT) has been widely used in pattern recognition thanks to its effectiveness and applicability for salient local structures. Its application for writer identification has been studied in many works. SIFT features can be encoded into a global vector or score, as in [37–39], or can be extracted directly for matching, as in [40].

Texture-based methods are efficient, and it is easy to compute texture features without binarization. However, high-contrast handwriting images are required to achieve better performance. These methods usually require more texture variations to model the writing style. They are also very time consuming and require a larger number of parameters and computations in the classification phase. This is mainly due to the high dimension of the extracted features. To deal with this issue, post-processing steps such as principal component analysis (PCA), linear discriminant analysis (LDA), and generalized discriminant analysis (GDA) are regularly used to transform the features into a low-dimensional representation space. In grapheme and contour approaches, binarization and segmentation techniques are very important to capture local information of the writing. When these techniques are poorly designed or applied to poor quality images, it is more difficult to capture the writing style. Therefore, overall performance is often limited when using these methods. Texture, contour, and grapheme approaches depend on features computed by algorithms and essentially designed by humans. Currently, these features can be learned automatically using deep learning. Deep learning indeed provides an easier way to obtain the desired features for the task at hand. Here we review some deep learning techniques related to our work. Deep learning for writer identification was first introduced by [41–43]. In all these approaches, a trained convolutional neural network (CNN) was used to generate features via the activation functions of the last fully connected layers. Fiel et al. [41] computed CNN features obtained by word and line segmentation from the IAM database. Christlein et al. [42] used CNN activations to capture writing features, which were

then encoded and matched with a Gaussian mixture model (GMM) supervector. The same authors in [43] trained LeNet and ResNet CNN models to learn the hidden layer activations. They used linear exemplar support vector machines (E-SVM) in classification to encode the learned features. Another interesting system presented in [44] trained ResNet-20 on the ICDAR2013 database to learn activation features that are PCA-whitened, encoded using the VLAD¹ method, and matched using the cosine distance metric. Chen et al. [45] introduced a semi-supervised feature learning based on a weighted label smoothing regularization (WLSR) method. In this method, a uniform distribution of labels is assigned to the additional unlabeled writing samples for data augmentation. Recently, Semma et al. [46] presented a writer identification system based on deep feature learning using a convolutional neural network. The overall system extracts points of interest from handwriting using FAST keypoints and Harris corner detector, uses the ConvNet model to detect features from these keypoint fields, and encodes the learned features using VLAD and triangulation embedding. The classification step is performed using the nearest neighbor rule.

Most deep learning-based SOTA methods train their models on either text fragments, words, or lines. In this way, given an input document with different font curves and shapes, the CNN model focuses only on the most discriminative parts of the input that dominate the training process, and the information in other parts is ignored. To consider all the discriminative parts of the document image at once, our approach uses both words and connected components as input to learn different features using different receptive CNN fields constructed in a two-stream CNN method. The combination of deep features learned from two different data instances helps the overall system to process multiple observations with different regions of interest, resulting in high identification performance. In the same context, He and Schomaker [47] presented a deep neural network consisting of two CNN streams, the fragment and the feature pyramid, to capture features from both fragment and word images. They concatenated the obtained features into a global feature representation, which is the input to the fully connected classification layer for writer identification. Similarly, Wang and Shiyu [48] proposed a global regular network consisting of two CNN streams to capture handwriting features from word-level and page-level images concatenated into a feature vector. Note that the classification process used in [47,48] is trained only on the depth concatenation of the features extracted from the multi-stream model. For example, in [47], given an input of M words and N segmented fragments, the model outputs only M feature vectors (each vector consists of the word vector concatenated with its corresponding fragments), which are then passed to the fully connected clas-

¹ Vector of Locally Aggregated Descriptors.

Table 1 Summary of well-known SOTA methods for offline text-independent writer identification

#Approach	#Category	#Feature + classifier	#Benchmark [writers/accuracy (%)]
Chawki et al. [20]	<i>Image-texture based methods</i>	(GLRL/GLCM) + NN	IFN [49] (130/82,62%)
Said et al. [21]		MGF + WED	Baseline dataset [21] (40/96%)
Bertolini et al. [22]		(LBP/LPQ) + SVM	BFL [50] (315/99,2%)—IAM [51] (650/96,7%)
Nicolaou et al. [23]		(SRLBP) + kNN	ICDAR13 [52] (250/98,5%)—CVL [53] (310/98,6%)
Singh et al. [24]		Handcrafted + (kNN/SVM)	IAM [51] (657/97,6%)—KHATT [54] (1000/95,6%)—Devnagri [24] (60/65,80%)
Chahi et al. [25]		CLGP + NN	IAM [51] (657/94,06%)—IFN [49] (411/98,54%)—CVL [53] (310/99,5%)
He and Schomaker [26]		(LBPruns/COLD) + NN	IAM [51] (650/89,9%)—Firemaker [55] (250/86,2%)
Bahram [27]		MLBP-IWSL + NN	IFN [49] (411/99,27%)—IAM [51] (657/98,17%)—CVL [53] (310/100%)
Chahi et al. [7]		(LPQ/LTP) + NN	IAM [51] (657/91,17%)—IFN [49] (411/97,81%)—CVL [53] (309/98,62%)
Bulacu and Schomaker [28]		Contour Hinge + NN	IAM [51] (650/81%)—Firemaker [55] (250/84%)
Siddiqi et al. [29]	<i>Contour-based methods</i>	(Chain/Polygone/Local) + NN	IAM [51] (650/79%)—RIMES [56] (375/78%)
Brink et al. [30]		Quill-Hinge + NN	IAM [51] (657/97%)—Firemaker [55] (250/86%)
He and Schomaker [31]		Delta-n Hinge + NN	IAM [51] (650/91,6%)—Firemaker [55] (250/89,2%)
Khalifa et al. [32]	<i>Grapheme-based methods</i>	Grapheme Codebook + SR-KDA	IAM [51] (650/92%)
Abdi and Khemakhem [8]		Beta-elliptic Model	IFN [49] (411/90,02%)
Ghiassi et al. [33]		(Curve/line) Codebook + NN	IAM [51] (650/94,8%)—Firemaker [55] (250/94,2)
He et al. [34]		Codebook + Junclets	IAM [51] (650/91,10%)—Firemaker [55] (250/89,80%)—CERUG [26] (105/85,7%)
He et al. [35]		(kSF/kCF) Codebooks	MPS [57] (143/64,3%)
Bulacu et al. [28]		Grapheme F4 + NN	IAM [51] (650/80%)—Firemaker [55] (250/75%)
Siddiqi et al. [29]		Codebook features + NN	IAM [51] (650/84%)—RIMES [56] (375/74%)
Ahmed et al. [36]		Codebook features + (NN/SVM)	IAM [51] (650/98,2%)—KHATT [54] (1000/89,5%)
Fiel et al. [58]		Caffenet + NN	ICDAR11 [59] (26/94,7%)—ICDAR13 [52] (250/88,5%)
Christlein et al. [42]	<i>Deep learning-based methods</i>	CNN + GMM	CVL [53] (310/99,4%)
Christlein et al. [43]		(ResNet/LetNet) + E-SVM	WI [60] (720/88,99%)
Christlein et al. [44]		ResNet-20 + E-SVM	CVL [53] (310/99,5%)—KHATT [54] (1000/99,6%)
Wang and Shiyu [48]		GRN network	CVL [53] (310/99,9%)
He and Schomaker [47]		FragNet network	CERUG-EN [26] (105/100%)—CVL [53] (310/99,1%)—IAM [51] (657/96,3%)
Semma et al. [46]		(Fast + Harris/ConvNet) + NN	IAM [51] (657/99,5%)—IFN [49] (411/99,8%)
Chen et al. [45]		CNN + WLSR	ICDAR13 [52] (250/97,7%)—CVL [53] (310/99,2%)

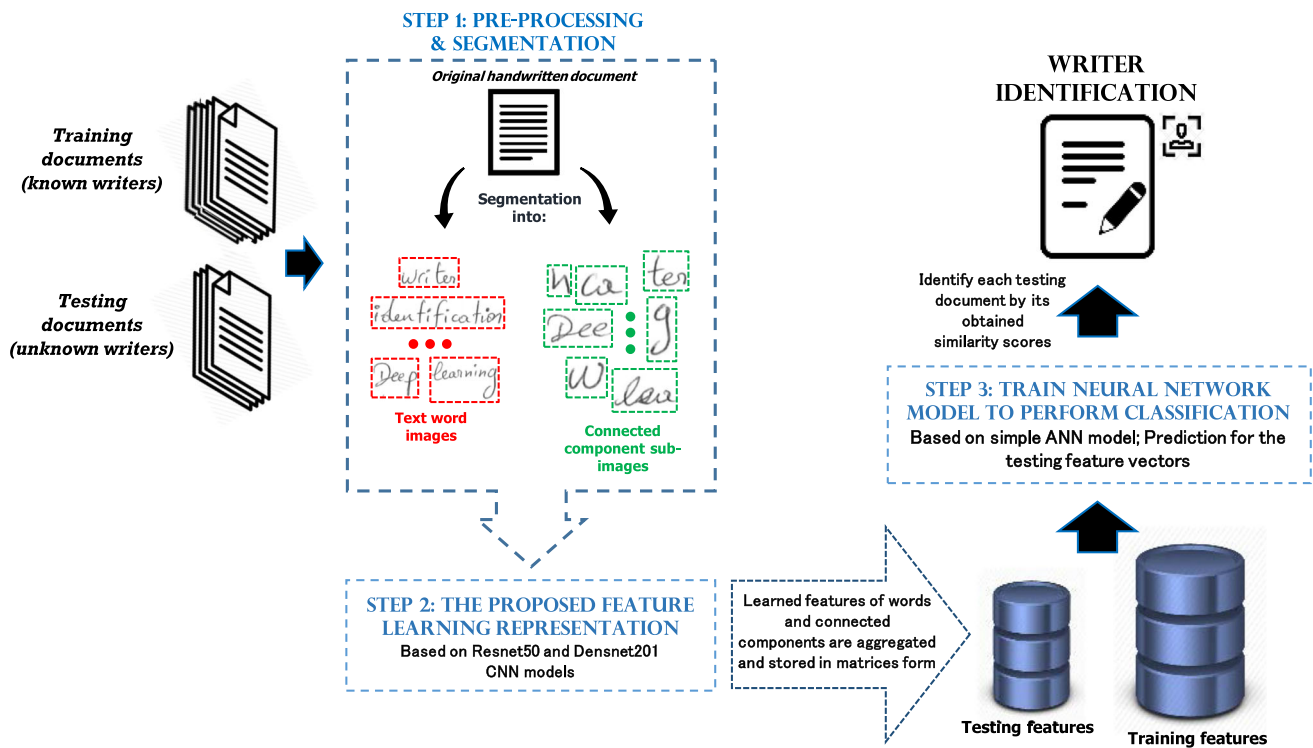


Fig. 1 Design and structure of the proposed *WriterINet* method

sifier to identify the writer. This concatenation strategy may lead to an unbalanced distribution of features, since the word images have more writing content compared to the fragments due to their small size. In our proposal, the ANN-classifier is learned on the feature vectors extracted from the two CNN streams, resulting in $(M + N)$ training feature vectors that allow enriching the ANN-learning process to improve the writer classification. With this strategy, we avoid the problem of unbalanced feature distribution, since each feature vector is considered as an observation and learned separately. Table 1 summarizes the above analyzed SOTA works and shows their results on SOTA benchmarks. Our method falls into the category of deep learning-based methods.

3 WriterINet architecture

This section presents the design and structure of the proposed *WriterINet* method for writer identification. As shown in Fig. 1, *WriterINet* consists of three steps: preprocessing of writing, feature extraction, and writer identification (classification). Each step is explained in more detail below.

3.1 Image preprocessing

Image preprocessing and segmentation is an important step to improve writer identification performance. As shown in

Fig. 1, each input handwritten sample is segmented into word images and related component sub-images. This step is very important because we need to isolate each word from the others and then separate the individual connected components in the writing. For word segmentation, we used the same scale space technique as in [61], which is based on the use of blob analysis and Gaussian filters. This segmentation algorithm is fast, easy to implement² and, more importantly, it ensures satisfactory segmentation results. As for the second segmentation, we extracted different regions of interest from the segmented word images, which are called connected components. This process is performed using a labeling function that places a small bounding box around each connected component, i.e., determines all connected adjacent pixels that form a contiguous trace. Each truncated contiguous component is identified by its specific class (writer). To quantify the writing content within each component, all connected neighboring pixels that have non-significant information such as diacritical marks and random writing traces are removed. To remove these unwanted details, either created during image segmentation or filled with the original background, a size-based thresholding technique is applied to the connected components using a small threshold σ ($\sigma < 50 \times 50$). An example of preprocessing and segmentation of a Chinese-English example is shown in Fig. 2.

² code: <https://github.com/githubharald/WordSegmentation>.

3.2 Feature learning

Feature extraction in writer identification is considered an important milestone that allows us to characterize the repetitive patterns of writing style. To extract such relevant features, a well-designed and effective feature extraction method is needed. If these features are poorly extracted from the writing, the system cannot perform correct writer identification even with a good classifier. At this stage, explicit features of the writing style must be extracted and descriptive details reflecting individual writing habits must be found. This complicates the task for SOTA feature extraction methods, which typically work with texture, gradient, and local shape information of the writing. To enable robust and reliable feature extraction, deep convolutional neural networks (CNNs) are used in this work as a powerful approach to model the variability within a writer and between different writers. Thanks to appropriate deep modeling and learning, the writing style can be automatically learned to obtain the desired and more representative features for writer identification. In this work, we propose a new deep learning-based approach called *WriterINet*, which consists of ResNet-50 [16] and DenseNet-201 [17] CNN models combined for feature extraction. These two networks are used simultaneously over two paths in one branch to enhance the identification performance. As shown in Fig. 1, *WriterINet* segments the input writing sample into word images and connected sub-images. To enrich the learning process, the words of the handwriting are the input of DenseNet-201, and the connected sub-images are the input of ResNet-50. We first trained DenseNet-201 and ResNet-50 models using word images and connected sub-images, respectively, using the training set of the IAM database. These pre-trained models are then used as a starting point for feature learning. Next, as shown in Fig. 3, we fine-tuned the two CNN models for each tested database by transfer learning to obtain two new models *Net1* and *Net2*. This is done by replacing the last layers with new layers adapted to the tested database. Since both the ResNet-50 and DenseNet-201 models require input images of size 224-by-224-by-3, we automatically resized the images before they were input to the two networks. After fine-tuning, we extracted the learned deep features from *Net1* and *Net2* and used these features to train an artificial neural network for classification, as shown in Fig. 4. As shown in Fig. 4, the *Net1* and *Net2* models construct a hierarchical representation of the input document image, extracting higher level features from deeper layers. Global average pooling layer activations of *Net1* and *Net2* are used to capture features of the handwritten input document. In this way, the learned features are pooled over all spatial locations, resulting in 2048 and 1920 bins in the feature dimension for each sub-image of the connected components by *Net1* and for each word image by *Net2*, respectively. In other words, for a handwritten input

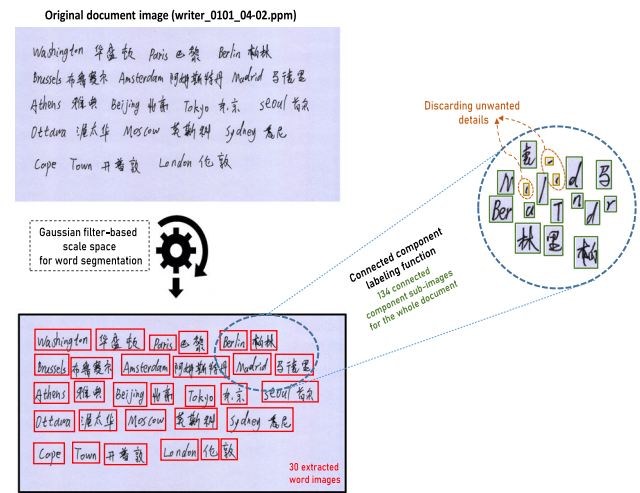


Fig. 2 Preprocessing of English-Chinese handwritten document taken from CERUG-MIXED [26] database

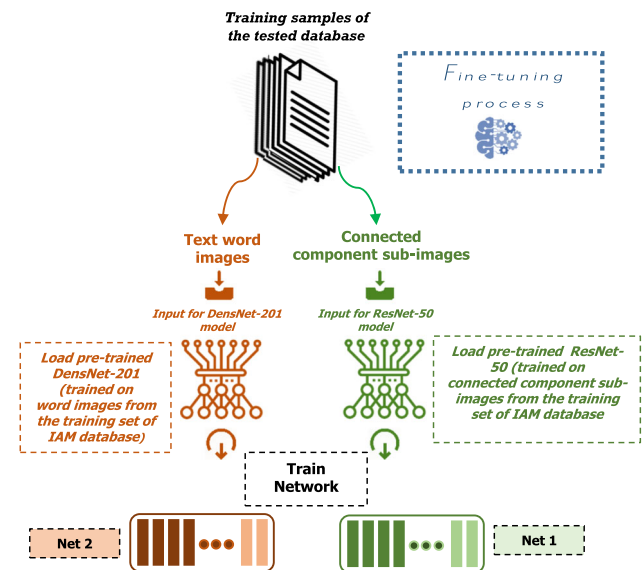


Fig. 3 Overview of the fine-tuning process

document, each extracted word image is characterized by its learned feature *Net2* (dimension 1920), and each extracted subimage with connected components is represented by its learned feature *Net1* (dimension 2048). *Net1* and *Net2* deep features of the same input document are arranged separately in matrix form. We zero-padded the features learned from *Net2* to obtain the same feature matrix size as the one derived from *Net1*. In this way, we can easily vertically concatenate the two feature matrices of *Net1* and *Net2* to form a holistic feature representation of the input document.

3.3 Writer identification process

At this level of our framework, we have learned features that characterize all extracted connected components and words

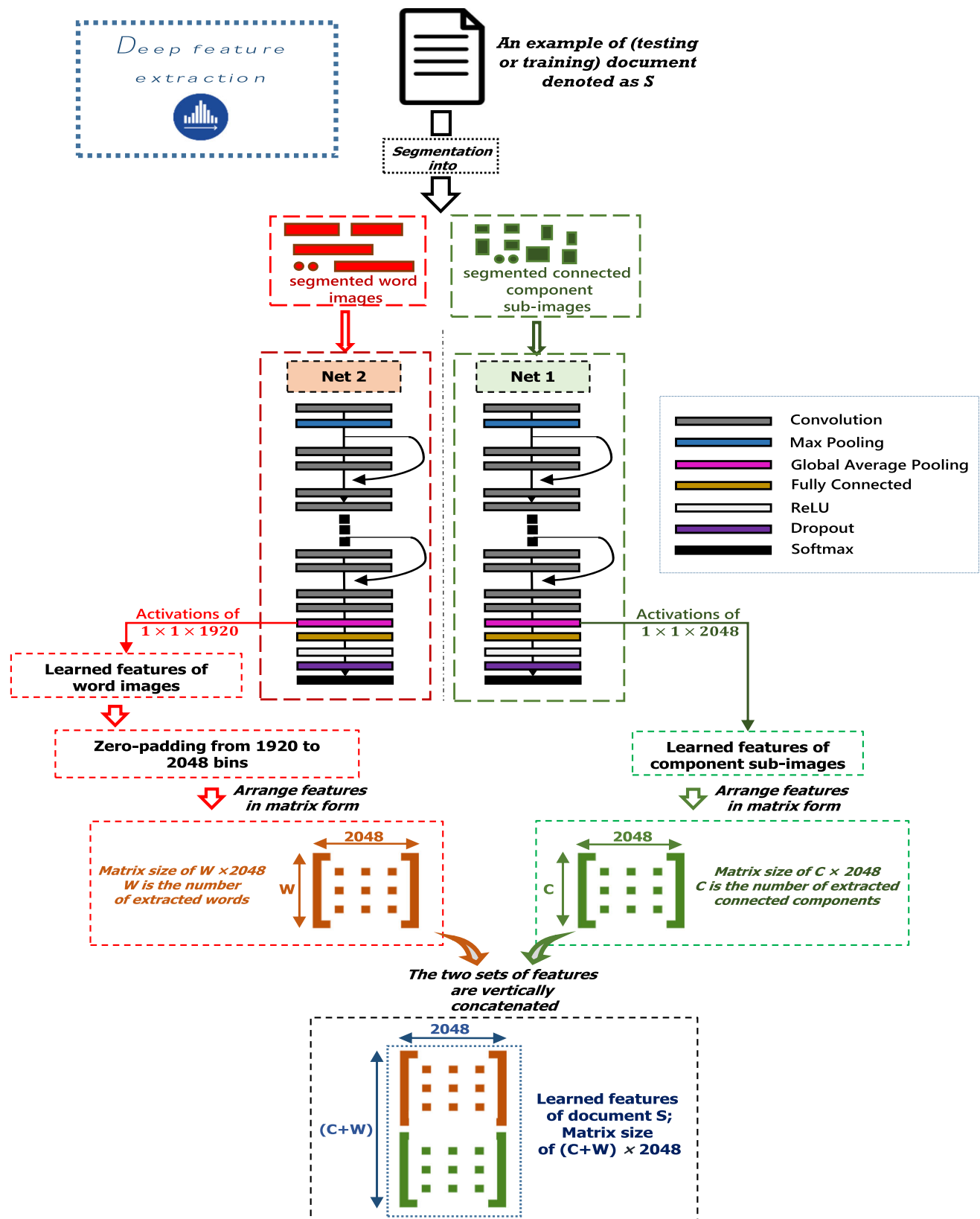


Fig. 4 Pipeline of deep feature extraction

from each writing sample. Note that for each sample, the number of extracted component sub-images and word images corresponds to the number of feature vectors representing the sample. To perform the identification (classification) process, as shown in Fig. 5, we trained a simple artificial neural network (ANN) on the learned features. This is achieved by building a simple architecture consisting of a 1D input, two fully connected layers, a softmax layer, and a classification layer. For a written query sample, the testing process is performed as follows (see Fig. 5): (i) following the methodology in Sect. 3.2, the learned features (represented by the feature matrix) corresponding to the segmented connected component sub-images and the word images are computed; (ii) the i th feature vector, i.e., the i th row of the feature matrix, is fed into the trained ANN-classifier to compute its similarity score vector Si , where $i = 1, \dots, F$ ($C + W$, the number of rows of the feature matrix, as shown in Fig. 4) denotes the number of segmented images (words and components), Si contains the similarity scores for the classes (writers), and its dimension is the number of writers (classes); (iii) calculate the final score of the j -th class by accumulating the scores obtained for all F feature vectors: $SCj = \frac{1}{F} \sum_{i=1}^F Si_j$, $j = 1, \dots, N$, N is the number of classes (writers); (iv) return the writer (j th class) with the highest similarity score SCj^* . Note that the similarity vector returned by the trained classifier ANN for each feature vector (row) of the feature matrix can be viewed as a probability distribution over all writers. Thus, we averaged the similarity vectors of all observations (segmented images, words, and connected components represented by the rows of the feature matrix) to produce the final prediction of the input writing sample. The test pipeline is shown in Fig. 5.

4 Evaluation

To validate the performance of our *WriterNet* framework, in this section we conduct extensive experiments on nine popular handwritten benchmarks in different written languages: (1) English CVL [53], (2) Arabic IFN/ENIT [49], (3) Dutch Firemaker [55], (4) hybrid language ICDAR2011 [59], (5) English IAM [51], (6) English-Chinese CERUG [26], and (7) English ICDAR2013 [52]. First, we describe the structure of the databases used, discuss and analyze the experimental results, present the sensitivity analysis of the number of neurons of the fully connected layer $FC1$ (cf. Sect. 3.3), and compare our system with the current SOTA for writer identification.

4.1 Experimental setup

For a fair performance comparison with SOTA, we set the standard evaluation protocols as in [39] for IAM and

ICDAR2011, in [62] for ICDAR2013 and IFN/ENIT, and in [47] for CERUG and Firemaker. As summarized in Table 2, the default configuration of each evaluated database is described as follows:

CVL This dataset has German–English documents produced by 310 scribes. 27 scribes were asked to complete 7 written documents: six in English and one in German. The remaining 283 scribes provided five writing samples, four in English and one in German. To obtain an evenly distributed database, only documents 1–5 from all writers were used in experiments with the following setup: 1 document (for testing)/4 documents (for training).

IFN/ENIT The Arabic IFN/ENIT database contains 411 writers. In the experiments, we used a maximum of 50 binary word images per writer. 60% of the word images are selected for the training set, and 40% are used for the tests.

IAM The proposed system is evaluated with a total set of 657 writers, each of whom provided a variable number of handwritten documents. In the experiments, we used the following two default configurations: # Document setup: a maximum of two documents are stored for each writer (one document is used for training and the other for testing). For writers who have contributed only one document, it will be split into two sub-pages. # Text-line setup: A maximum of 14 text lines are used for each author. 60% of the data is used for training, and the remaining 40% is used for testing. Note that we used the training set of this database to train *Net1* and *Net2* CNN models for feature extraction (cf. Fig. 3).

Firemaker The Dutch Firemaker database includes 250 writers, each of whom contributed to create document samples with handwritten text of varying content scanned at 300 dpi/grayscale. The writers were asked to fill four different A4 pages with handwritten text. Page 1 contains five short paragraphs in normal handwriting, i.e., lowercase letters with some capital letters at the beginning of sentences and names. On page 2, two paragraphs were handwritten in capital letters only. On page 3, writers provided fake text, while on page 4 they were asked to reproduce the content of a given cartoon in their own handwritten words. In our experimental study, we selected only two pages: Page 4 is used for testing, while Page 1 is used for training.

ICDAR2011 The ICDAR2011 database consists of English, French, Greek, and German document samples scanned at 300 dpi and collected from 26 writers. Each writer was asked to complete a total of 8 pages, i.e., two full pages of text for each language. For our experiments, we used a subset of ICDAR2011, called the CICDAR2011 dataset, in which only the first two lines of text on each page were truncated. For each author, the first five pages are the training dataset and the other three pages are the test dataset.

ICDAR2013 The ICDAR2013 benchmark database contains Greek and English document samples with handwritten text of variable content scanned in binary format and pro-

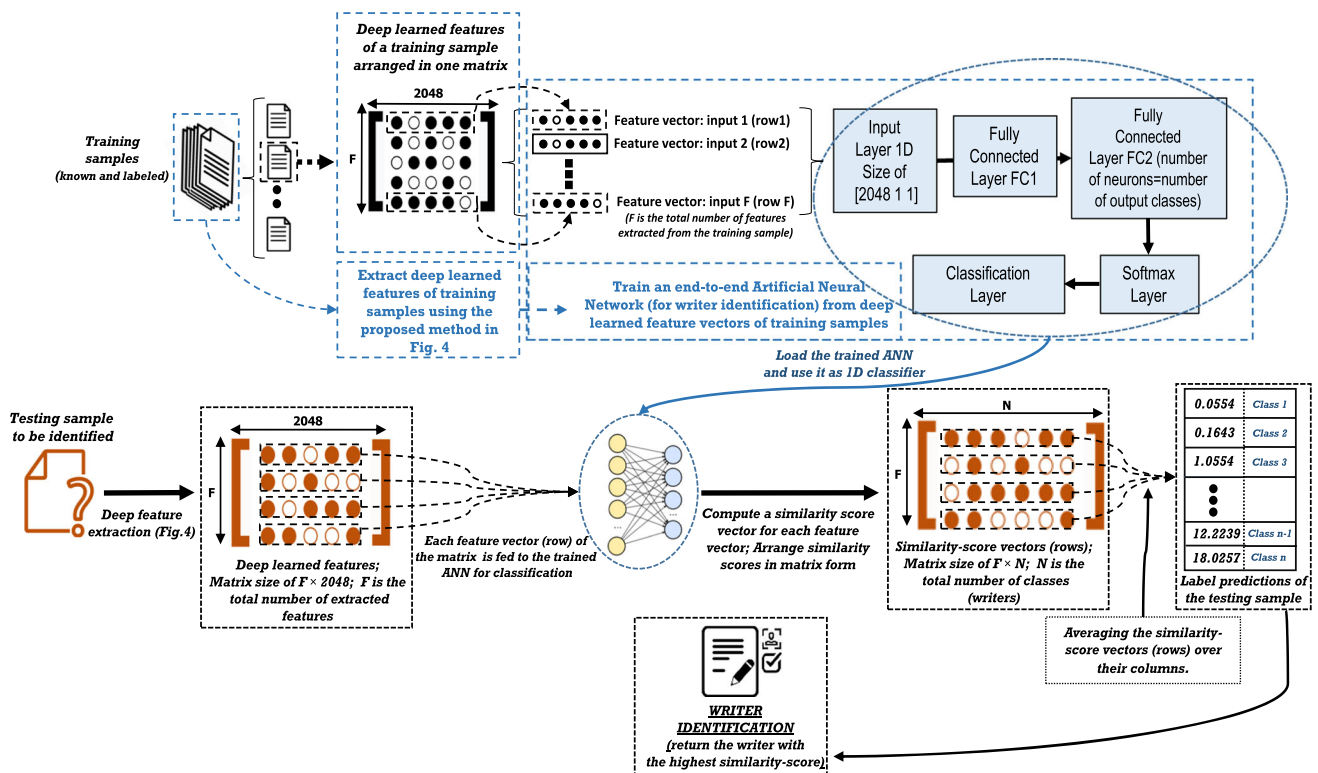


Fig. 5 Classification process

Table 2 Description of the tested databases

Benchmark	Language	Writers (classes)	Input text for testing
CERUG-CN	Chinese	105	1 Document
CERUG-EN	English	105	1 Document
CERUG-MIXED	Chinese/English	105	1 Document
CVL	English/German	310	1 Document
IFN/ENIT	Arabic	411	20 Words at most
IAM (document setup)	English	657	1 Document
IAM (text-line setup)	English	657	About 6 text lines
Firemaker	Dutch	250	1 Document
ICDAR2011	English/French	26	3 Documents
	Greek/German		
ICDAR2013	Greek/English	250	2 Documents

The use of 3 or 2 documents in testing is more difficult than using 1 document

duced by 250 writers. In our experiments, four documents (two in English and two in Greek) are used per scribe. A half-half setup is performed for this database, where two documents are used for testing and the others for training.

CERUG The benchmark CERUG consists of English and Chinese handwritten documents scanned at 300 dpi, 8 bits/pixel, grayscale, and captured by 105 writers. Four different A4 documents are available per writer. Two pages in Chinese, one page in English, and the fourth page is mixed. The English page was divided into two sub-pages, each containing one paragraph. This forms the subset known

as CERUG-EN. In the same way, the English-Chinese page has been split into two sub-pages to form the dataset CERUG-MIXED. The subset containing the two pages in Chinese is known as CERUG-CN. For the three subsets, we chose the following half-half setup: One image is used for testing and the other for training.

4.2 Implementation details

Experiments were run on two Alienware Aurora R8s with Core i9-9900K-9th Processor 5GHz Boost, 16 threads, two

GPU NVIDIA -RTX2080-8GB and 64 GB RAM. As for the training options of *Net1* and *Net2* CNN networks, we set (1) training solver: Adam's algorithm proposed in [63] was used as the optimizer for training *Net1* and *Net2*; (2) initial learning rate: In the training process, we *freeze* the weights of previous layers in *Net1* and *Net2* networks by setting the learning rate in these layers to zero. This significantly speeds up network training and prevents the frozen layers from overfitting to the new handwriting database during transfer learning (fine tuning). Therefore, we set the learning rate to zero in the first 24 and 90 layers for *Net1* and *Net2* networks, respectively. Next, we set the initial learning rate to a small value of 0.0001 (in both *Net1* and *Net2*) to slow down learning in the transferred layers that are not yet frozen; (3) training cycle: To update the weights of *Net1* and *Net2*, we set 8 as the maximum number of epochs and 60 as the mini-batch size for each training iteration; (4) validation frequency and execution environment: 14 is set as the number of iterations between validation metric evaluations. For hardware setup, we used two GPUs per machine via a local parallel pool.

4.3 Results

To evaluate our *WriterINet*, we applied the strategy proposed in Sect. 3.3 to obtain the writer identification rate expressed in %. This value is defined as the number of correctly classified samples relative to the total number of test instances across all writers. To investigate the performance stability and effectiveness of the proposed *WriterINet*, we evaluate how the number of neurons in the fully connected layer *FC1* (cf. Fig. 5) affects the overall performance. The number of neurons is set to 512, 1024, 2048, and 4096 as indicated in Table 3, and the identification accuracy for each setting is recorded across all benchmarks evaluated. Based on the results reported in Table 3, it is clear that the highest performance is achieved for the IFN/ENIT, ICDAR2013, CERUG-CN, CERUG-EN and CERUG-MIXED databases when 512 is set as the number of neurons of the fully connected layer *FC1*. The number of neurons of 1024 experimentally proves to be the best configuration for CVL, ICDAR2013 and CERUG-EN databases with the highest score of 100%. The configuration (*FC1*-2048) gives a SOTA performance of 99.54% and 98.4% for the IAM (text-line setup) and Firemaker databases, respectively. For the ICDAR2011 database, all tested configurations of *FC1* allow reaching 100%.

As described in Sect. 3, our approach consists of two CNN models, ResNet-50 (*Net1*) and DenseNet-201 (*Net2*), which are combined in a hierarchical representation to learn different and complex writing patterns for better characterization of writing style. The learned features constructed from *Net1* and *Net2* are fed to an artificial neural network (ANN)

Table 3 The impact of the number of neurons of the fully connected *FC1* on the system performance over all tested benchmarks

	IFN/ENIT		CVL		ICDAR2013	
	Top-1	Top-2	Top-1	Top-2	Top-1	Top-2
<i>FC1</i> -512	99.75	100	99.67	100	99	99.6
<i>FC1</i> -1024	99.27	99.75	100	100	99	99.6
<i>FC1</i> -2048	99.02	99.51	99.03	100	98.6	99
<i>FC1</i> -4096	98.54	99.27	97.09	97.74	98	98.6
	IAM		Firemaker		ICDAR2011	
	Top-1	Top-2	Top-1	Top-2	Top-1	Top-2
<i>FC1</i> -512	99.39	99.69	98	99.2	100	100
<i>FC1</i> -1024	99.39	99.69	98.4	99.6	100	100
<i>FC1</i> -2048	99.54	99.54	98.4	99.6	100	100
<i>FC1</i> -4096	99.23	99.54	97.6	98.4	100	100
	CERUG-CN		CERUG-EN		CERUG-MIXED	
	Top-1	Top-2	Top-1	Top-2	Top-1	Top-2
<i>FC1</i> -512	95.24	99.04	100	100	100	100
<i>FC1</i> -1024	93.33	97.14	100	100	99.04	100
<i>FC1</i> -2048	92.38	94.28	98.09	100	97.14	99.04
<i>FC1</i> -4096	90.47	92.38	95.24	98.09	96.19	98.09

for writer identification. To validate the effectiveness and superiority of our system, three different combinations of components are tested. First, only the *Net1* model is used for feature learning along with ANN for classification. In the second scenario, we use the same strategy as in the first scenario and replace only *Net1* with *Net2*. The third configuration (*WriterINet*) examines the effectiveness of both *Net1* and *Net2* in extracting more discriminative features along with ANN as the classifier. Given a handwritten input document, the *Net1* model processes the sub-images of the connected components cut from the handwritten words, while *Net2* processes the word images segmented from the document. Table 4 shows the classification results obtained with the three combinations of system components. It is worth noting that the optimal parameter values for the number of neurons in the fully connected layer *FC1* of the model ANN (see Table 3) are used to obtain the classification results reported in Table 4. As shown in Table 4, the *WriterINet* model achieves the highest performance compared to the other configurations in all benchmarks tested. For example, in the Firemaker, CERUG-CN and CERUG-MIXED datasets, *WriterINet* provides a performance improvement of +2%(Firemaker)/+4.77%(CERUG-CN)/+2.86%(CERUG-MIXED) and +2.4%(Firemaker)/+2.86%(CERUG-CN)/+1.91%(CERUG-MIXED) over the (*Net1* + ANN) and (*Net2* + ANN) models, respectively. This proves the reliability of the combination of the two CNN streams in learning more discriminative features from hand-

Table 4 Performance results of the different configurations of the proposed system

#IFN/ENIT			#ICDAR2013			#IAM		
<i>Net1 + ANN</i>	<i>Net2 + ANN</i>	<i>WriterINet</i>	<i>Net1 + ANN</i>	<i>Net2 + ANN</i>	<i>WriterINet</i>	<i>Net1 + ANN</i>	<i>Net2 + ANN</i>	<i>WriterINet</i>
97.81	97.32	99.75	97.4	97.8	99	98.63	98.47	99.54
#Firemaker			#CVL			#ICDAR2011		
<i>Net1 + ANN</i>	<i>Net2 + ANN</i>	<i>WriterINet</i>	<i>Net1 + ANN</i>	<i>Net2 + ANN</i>	<i>WriterINet</i>	<i>Net1 + ANN</i>	<i>Net2 + ANN</i>	<i>WriterINet</i>
96.40	96	98.4	98.71	98.38	100	98.71	98.71	100
#CERUG-CN			#CERUG-MIXED			#CERUG-EN		
<i>Net1 + ANN</i>	<i>Net2 + ANN</i>	<i>WriterINet</i>	<i>Net1 + ANN</i>	<i>Net2 + ANN</i>	<i>WriterINet</i>	<i>Net1 + ANN</i>	<i>Net2 + ANN</i>	<i>WriterINet</i>
90.47	92.38	95.24	97.14	98.09	100	98.09	99.05	100

writing data of different forms, consisting of both connected components and word images (multi-observation), to enrich the learning process of the subsequent ANN model in correctly classifying query writers.

4.4 Performance comparison with the current state of the art

Although several studies have addressed writer identification, this problem remains challenging due to the wide variability of writing. Recent advances in pattern recognition and machine learning have enabled researchers to develop new frameworks and reliable approaches that would yield higher performance in writer identification. In this section, a comprehensive performance evaluation is conducted to compare the proposed *WriterINet* system with existing systems in the literature. Only known published studies that used the total number of classes in the ICDAR2013, CERUG, IFN/ENIT, IAM, Firemaker, ICDAR2011, and CVL databases are considered in this evaluation. Tables 5, 6, and 7 reflect the top 1 identification results from *WriterINet* as well as those from the current literature for the nine benchmarks tested. To provide a meaningful and informative comparison of performance, we set up the databases setup as settled in the nearest best approaches for each database evaluated (see Sect. 4.1).

Performance on IFN/ENIT (cf. Table 5) # In this dataset, the proposed *WriterINet* outperforms all the SOTA approaches studied with a top 1 score of 99.75%. The texture-based system presented in [27] ranks second with an accuracy of 99.27%. Our previous works [7,9,25,62] achieved lower classification rates than *WriterINet* (by about 1.21–2.19%).

Performance on CVL, ICDAR2013, CERUG-EN, CERUG-MIXED, and ICDAR2011 (see Tables 5, 6, 7) # It is clear that most of the SOTA approaches tested in these benchmarks are more or less competitive. The texture method presented in [27] and our proposal are the best performing systems in the ICDAR2013 dataset, with peak accuracy of 100%

and 99% respectively. Remarkably, our *WriterINet* system can perfectly discriminate all classes with 100% accuracy in the CVL, CERUG-EN, CERUG-MIXED and ICDAR2011 datasets. This high performance confirms the reliability and validity of our framework for effective query writer identification.

Performance on IAM (cf. Table 6) # The proposed *WriterINet* provides an efficient and reliable solution to accurately identify the queried writers. Compared to our previous work, *WriterINet* achieves a significant improvement with a maximum score of 99.54% (text-line setup), outperforming all other previously studied approaches. In document setup, the proposed method and the work presented in [27] achieve a score of 98.17%, which is comparable to the performance achieved by the first performing system in [76] (score of 98.5%). However, our system outperforms the same system in [76] on the Firemaker and ICDAR2011 databases by 6% and 4.8%, respectively (see Table 6). In document setup, the proposed method and the work presented in [27] achieve a score of 98.17%, which is comparable to the performance of the first-rank system in [76] (score of 98.5%). However, our system outperforms the same system in [76] on the Firemaker and ICDAR2011 databases by 6% and 4.8%, respectively (see Table 6). Note that *WriterINet* and the method presented in [36] have competing performances, with advantage of our proposal since the system in [36] used only 650 writers, while we evaluated our approach on a total of 657 writers from the benchmark tested.

Performance on Firemaker (cf. Table 6) # The experimental results show that our *WriterINet* and the systems in [27,69] are the best performing systems with top-1 identification results of 98.4% and 99.6%, respectively. However, our approach outperforms the same work in [69] in CVL and IAM databases by 3.84% and 2.04%, respectively. It also outperforms the texture-based approach in [27] in the IFN/ENIT dataset by 0.48%.

Performance on CERUG-CN (cf. Table 7) # This benchmark shows that SOTA methods fail in characterizing Chinese characters. Our *WriterINet* approach performs better

Table 5 Comparison with state-of-the-art results on IFN/ENIT, CVL, and ICDAR2013 databases

System	#Year	#Language	#Database	#Writers	#Feature + classifier	Writer identification Top-1 score (%)
Hannad et al. [64]	2016	Arabic	IFN/ENIT	411	LPQ + Hamming	94.89
Khan et al. [39]	2019	Arabic	IFN/ENIT	411	SIFT and RootSIFT with GMM	97.28
Kumar and Sharma [65]	2019	Arabic	IFN/ENIT	411	DCWI + SVM and SBC	97.50
Chahi et al. [9]	2018	Arabic	IFN/ENIT	411	BWLBC + Hamming	97.56
Hadjadji and Chibani [66]	2018	Arabic	IFN/ENIT	411	LPQ, RL, and oBIF + OC-K-Means	97.56
Chahi et al. [7]	2019	Arabic	IFN/ENIT	411	LPQ and LTP + NN-Hamming	97.81
Kumar and Sharma [67]	2020	Arabic	IFN/ENIT	411	CNN	98.24
Chahi et al. [62]	2020	Arabic	IFN/ENIT	411	LSTP + NN-Hamming	98.28
Chahi et al. [25]	2020	Arabic	IFN/ENIT	411	CLGP + NN-Hamming	98.54
Bahram [27]	2022	Arabic	IFN/ENIT	411	MLBP-IWSL + NN	99.27
Proposed WriterNet		Arabic	IFN/ENIT	411	CNN + ANN	99.75
Kessentini et al. [68]	2018	English and German	CVL	310	SVM with DST	94.83
Javidi and Jampour [69]	2020	English and German	CVL	310	CNN + HTD	96.16
Fiel and Sablatnig [58]	2015	English and German	CVL	309	CNN + Nearest Neighbor	98.9
Khan et al. [39]	2019	English and German	CVL	310	SIFT and RootSIFT with GMM	99.03
He and Schomaker [47]	2020	English and German	CVL	310	CNN	99.1
Chen et al. [45]	2019	English and German	CVL	310	CNN-WLSR	99.2
He et al. [70]	2021	English and German	CVL	310	GR-RNN	99.3
Christlein et al. [42]	2015	English and German	CVL	310	CNN with GMM	99.4
Kumar and Sharma [67]	2020	English and German	CVL	310	CNN	99.35
Christlein and Maier [44]	2018	English and German	CVL	310	CNN + E-SVM	99.5
Tang and Wu [71]	2016	English and German	CVL	310	CNN + Joint bayesian	99.7
Mohammed et al. [40]	2017	English and German	CVL	310	FAST keypoints + Local NBNN	99.8
Chahi et al. [9]	2018	English and German	CVL	309	BWLBC + Hamming	98.38
Chahi et al. [7]	2019	English and German	CVL	309	LPQ and LTP + NN-Hamming	98.62
Chahi et al. [25]	2020	English and German	CVL	310	CLGP + NN-Hamming	99.51
Chahi et al. [62]	2020	English and German	CVL	310	LSTP + NN-Hamming	99.83
Bahram [27]	2022	English and German	CVL	310	MLBP-IWSL + NN	100

Table 5 continued

System	#Year	#Language	#Database	#Writers	#Feature + classifier	Writer identification Top-1 score (%)
Proposed <i>WriterINet</i>		English and German	CVL	310	CNN + ANN	100
Fiel and Sablatnig [58]	2015	English and Greek	ICDAR2013	250	CNN + Nearest Neighbor	88.5
CVL-IPK method. [52]	2013	English and Greek	ICDAR2013	250	Fisher Vector + Cosine	90.9
TEBESSA-c method. [52]	2013	English and Greek	ICDAR2013	250	Hinge with Run-length + Manhattan	93.4
HIT-ICG method. [52]	2013	English and Greek	ICDAR2013	250	SIFT + Chi-square	94.8
CS-UMD-b method. [52]	2013	English and Greek	ICDAR2013	250	Contour gradient + K-means	95
Christlein et al. [72]	2014	English and Greek	ICDAR2013	250	RootSIFT with GMM	97.1
Chen et al. [45]	2019	English and Greek	ICDAR2013	250	CNN-WLSR	97.7
Chahi et al. [62]	2020	English and Greek	ICDAR2013	250	LSTP + NN-Hamming	98.4
Bahram [27]	2022	English and Greek	ICDAR2013	250	MLBP-JWSL + NN	100
Proposed <i>WriterINet</i>		English and Greek	ICDAR2013	250	CNN + ANN	99

Table 6 Comparison with state-of-the-art results on IAM, Firemaker, and ICDAR2011 databases

System	#Year	#Language	#Database	#Writers	#Feature + Classifier	Writer identification Top-1 score (%)
Hannad et al. [64]	2016	English	IAM (text-line setup)	657	LPQ + Hamming	89.54
He and Schomaker [26]	2017	English	IAM (document setup)	650	COLD-LBPruns + Nearest Neighbor	89.90
Chahi et al. [9]	2018	English	IAM (text-line setup)	657	BWLBC + Hamming	90.11
Nguyen et al. [73]	2019	English	IAM (document setup)	650	CNN	90.12
He et al. [34]	2015	English	IAM (document setup)	650	Junclets	91.10
Chahi et al. [7]	2019	English	IAM (text-line setup)	657	LPQ and LTP + NN-Hamming	91.17
Durou et al. [74]	2019	English	IAM	650	OBI and Grapheme + K-nearest neighbor	92
Hadjadji and Chibani [66]	2018	English	IAM (text-line setup)	657	LPQ, RL, and oBIF + OC-K-Means	94.51
He and Schomaker [47]	2020	English	IAM	657	CNN	96.3
He et al. [70]	2021	English	IAM	657	GR-RNN	96.4
Litifu et al. [75]	2021	English	IAM	657	DF-ANOVA	96.9
Kumar and Sharma [67]	2020	English	IAM (document setup)	657	CNN	97.27
Javidi and Jampour [69]	2020	English	IAM	657	CNN + HTD	97.50
Kumar and Sharma [65]	2019	English	IAM (text-line setup)	657	DCWI + SVM and SBC	97.80
Khan et al. [39]	2019	English	IAM (document setup)	650	SIFT and RootSIFT with GMM	97.85
Chahi et al. [25]	2020	English	IAM (text-line setup)	657	CLGP + NN-Hamming	94.06
Chahi et al. [62]	2020	English	IAM (text-line setup)	657	LSTP + NN-Hamming	96.80
Ahmed et al. [36]	2022	English	IAM (document setup)	650	Codebook features + (NN/SVM)	98.2
Bahram [27]	2022	English	IAM (document setup)	657	MLBP-IWSL + NN	98.17
Wu et al. [76]	2014	English	IAM (document setup)	657	SDS + SOH	98.5
Proposed WriterINet		English	IAM (text-line setup)	657	CNN + ANN	99.54
Proposed WriterINet		English	IAM (document setup)	657	CNN + ANN	98.17
He and Schomaker [26]	2017	Dutch	Firemaker	250	COLD-LBPruns + Nearest Neighbor	86.2
He et al. [34]	2015	Dutch	Firemaker	250	Junclets	89.80
Nguyen et al. [73]	2019	Dutch	Firemaker	250	CNN	92.38
Wu et al. [76]	2014	Dutch	Firemaker	250	SDS + SOH	92.4
Litifu et al. [75]	2021	Dutch	Firemaker	250	DF-ANOVA	96.4
He and Schomaker [47]	2020	Dutch	Firemaker	250	CNN	97.6
Khan et al. [39]	2019	Dutch	Firemaker	250	SIFT and RootSIFT with GMM	97.98
Chahi et al. [25]	2020	Dutch	Firemaker	250	CLGP + NN-Hamming	97.60
Chahi et al. [62]	2020	Dutch	Firemaker	250	LSTP + NN-Hamming	98
He et al. [70]	2021	Dutch	Firemaker	250	GR-RNN	98.8
Javidi and Jampour [69]	2020	Dutch	Firemaker	250	CNN + HTD	99.6
Bahram [27]	2022	Dutch	Firemaker	250	MLBP-IWSL + NN	99.6

Table 6 continued

System	#Year	#Language	#Database	#Writers	#Feature + Classifier	Writer identification Top-1 score (%)
Proposed WriterINet		Dutch	Firemaker	250	CNN + ANN	98.4
Louloudis et al. [59]	2011	Hybrid	ICDAR2011	26	GLRL + Manhattan	87.50
Louloudis et al. [59]	2011	Hybrid	ICDAR2011	26	GMSF + Variance Weighted Chi-square	90.90
Fiel and Sablatng [41]	2015	Hybrid	ICDAR2011	26	CNN + Nearest Neighbor	94.7
Wu et al. [76]	2014	Hybrid	ICDAR2011	26	SDS + SOH	95.20
Mohammed et al. [40]	2017	Hybrid	ICDAR2011	26	FAST keypoints + Local NBNN	98.6
Khan et al. [39]	2019	Hybrid	ICDAR2011	26	SIFT and RootSIFT with GMM	100
Chahi et al. [25]	2020	Hybrid	ICDAR2011	26	CLGP + NN-Hamming	100
Proposed WriterINet		Hybrid	ICDAR2011	26	CNN + ANN	100

Table 7 Comparison with state-of-the-art results on CERUG-CN, CERUG-EN, and CERUG-MIXED databases (105 writers)

System	#Year	#Writers	CERUG-CN (%)	CERUG-EN (%)	CERUG-MIXED (%)
Hinge [28]	2007	105	90.8	12.3	84.7
Quill [30]	2012	105	82.7	15.8	74.8
Junclets [34]	2015	105	90.4	87.1	85.7
COLD + LBPruns [26]	2017	105	93.8	97.1	98.5
FragNet-64 [47]	2020	105	–	100	–
GR-RNN [70]	2021	105	–	99.1	–
Proposed WriterINet		105	95.24	100	100

with a classification rate of 95.24%. In light of the above results and discussion, *WriterINet* proves to be: (1) practical and efficient in writing characterization; (2) flexible and robust to character shape variations with complex structures; (3) reliable and language independent in handling seven different challenging languages (English, Arabic, French, German, Chinese, Dutch, Greek, and hybrid language).

5 Conclusion

This work addresses text-independent offline recognition of handwritten documents. We proposed an effective deep learning-based approach called *WriterINet*. The first step of *WriterINet* aimed to segment each written document into word images and related sub-images. Words were segmented using a scale space technique based on blob analysis and Gaussian filters. Connected components were collected using labeling-based bounding box technique to detect connected neighboring pixels, i.e., contiguous writing traces. At this level, the connected components and words were fed into our proposed deep CNN architecture, which combines two fine-tuned CNN models derived from ResNet-50 and DenseNet-201 models, for feature learning. In the classification phase, an efficient and simple 1D artificial neural network (ANN) model is built to classify the learned features for writer identification. Our *WriterINet* framework delivered high peak performance on nine public handwritten writer identification datasets: Arabic IFN /ENIT, English ICDAR2013, English CVL, hybrid language ICDAR2011, English IAM, Dutch Firemaker, English CERUG-EN, Chinese CERUG-CN, and English-Chinese CERUG-MIXED.

In future work, we plan to investigate the ability of *WriterINet* for text-independent verification and gender classification. We also plan to propose the collection of a new and large mixed handwritten benchmark with different languages for writer identification. This will make it possible to provide a unique database that combines all challenges for evaluating research on writer identification.

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